



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 2D: Example of VHDL Process

Process Anatomy

```
proc_name: process (sensitivity list)
    declarations;
begin
    statements;
end process;
```

variables valid inside the process (optional)

```
PROCESS (A, B, Ci)
BEGIN
    S <= A XOR B XOR Ci;
    Co <= (A AND B) OR (B AND Ci) OR (A AND Ci);
END PROCESS;
```

- A process is contained inside of an architecture

Process Example

```
-- entity omitted for simplicity
architecture behavioral of process_example is
    signal A, B, Ci, S, Co : std_logic;
begin

    process (A, B, Ci)
        -- variables, if any, go here
    begin
        S <= A XOR B XOR Ci;
        Co <= (A AND B) OR (B AND Ci) OR (A AND Ci);
    end process;

end behavioral;
```

Variables

- Variables are for storing (an) intermediate computation(s) in the middle of a process
- Good for organizational purposes

```
process (A, B, Ci)
```

```
    variable var_s1, var_s2, var_s3 : std_logic;
```

```
    begin
```

```
        S <= A XOR B XOR Ci; ←
```

signals visible outside
process (defined elsewhere)

NOT visible
outside process

```
        { var_s1 := A AND B;  
          var_s2 := B AND Ci;  
          var_s3 := A and Ci;
```

visible outside
process →

```
        Co <= var_s1 OR var_s2 OR var_s3;
```

```
    end process;
```

Signals vs. Variables

- A signal defined at the architecture level is visible outside the process
- Variables are only visible inside the process they are used in
- Signals are used to communicate between processes/CSA's and components, within the rest of the architecture
- In general:
 - Use signals for values that go outside the process, and use variables to compute values within a process
 - Use variables to breakdown complicated statements; use signals to hold the final value

Signals vs. Variables Continued

